



A Novel Proposal For Mammogram Mass Segmentation Using an Improved Fuzzy C-means

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Abstract

Given the high global incidence of breast cancer, developing early and accurate diagnostic methods for mammography is increasingly urgent. In this paper, we propose a new method (MoFCM), which is improved from the Fuzzy C-Means (FCM) algorithm by applying fuzzy logic techniques during image preprocessing, thereby preserving important features and enhancing segmentation efficiency. In addition, the proposed method integrates spatial information and spatial coefficients to help the FCM algorithm improve clustering ability in cases where noise appears in the input image. To evaluate the effectiveness of the proposed method, we conduct experiments, comparing it with the FCM and Enhanced Fuzzy C-Means (EnFCM) algorithms through clustering and segmentation efficiency evaluation indexes. The experimental results demonstrate that the proposed method achieves accurate mass segmentation efficiency on the DDSM, MVinDr and our datasets with the highest accuracy of 94.7%. This research contributes to the field of computer vision in medicine, paving the way for intelligent solutions to support breast cancer diagnosis, improve treatment efficiency, and enhance the quality of healthcare.

Keywords Fuzzy clustering · Fuzzy C-means · Mammogram

Introduction

Breast cancer poses a significant risk to women's health. According to GLOBOCAN [1] 2020 statistics, the incidence and mortality of cancer are increasing globally. In Vietnam, there were approximately 182,563 new cancer cases and 122,690 cancer-related deaths reported. Out of every 100,000 people, 159 were diagnosed with cancer and 106 died from the disease. Drug reactions associated with

different breast cancer subtypes have clear implications for treatment outcomes; therefore, accurate classification is very important. Breast cancer, like other types of cancer, is a type of malignant tumor that appears in the chest and breast area of the patient. Although breast cancer primarily affects women, it can also occur in men, albeit rarely. This is a common malignancy, but can be cured if detected early, with about 80% of patients recovering completely in the early stages. Most cases originate from the milk ducts, while others begin to develop from the milk sacs or lobules. Breast cancer or any cancer can be life-threatening if not detected and treated promptly. If left untreated, the tumor can spread to the bones or other vital organs, causing pain and endangering life. Mammography is a method that uses low-dose X-rays to record two-dimensional images of breast tissue, supporting early detection and screening for breast cancer. This technique allows assessment of breast tissue density, detection of abnormal structures such as microcalcifications, tumors, or lesions of mammary gland tissue at a very early stage, when there are no clinical symptoms. Mammography works based on differences in X-ray absorption between tissue components. Fatty tissue absorbs less X-rays and appears darker (dark), while dense tissue

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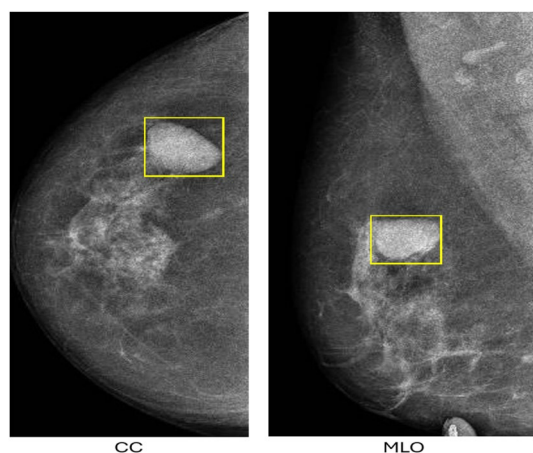


Fig. 1 Two standard views: CC (cranio-caudal) and MLO (medio-lateral oblique)

and lesions absorb more X-rays and appear brighter (white) on the image. The scan usually involves two standard views: CC (cranio-caudal) and MLO (medio-lateral oblique) view (Fig. 1), which ensure full visualization of the breast structure. Modern systems also support 3D imaging techniques (digital breast tomosynthesis (DBT), which reconstructs cross-sectional images and reduces tissue overlap.

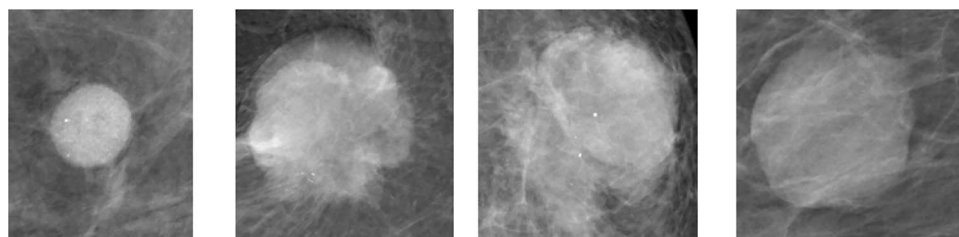
Mammography is commonly used for screening purposes - applied to asymptomatic women, and diagnostic purposes when there are suspicious clinical signs. The World Health Organization (WHO) and many medical associations recommend that women aged 40 and over should have regular mammography every 1–2 years depending on their risk level. Large-scale clinical studies have shown that implementing a regular mammography screening program reduces breast cancer mortality by 20–40%, while increasing the rate of early detection. However, diagnostic efficiency is still affected by image quality and breast tissue characteristics, especially in women with dense breast tissue. In this context, the need to apply artificial intelligence diagnostic support systems is increasing, to minimize errors, increase consistency and improve screening efficiency. Although mammography is a popular and highly effective method of breast cancer screening for early detection, it has some technical and clinical limitations. These challenges can affect diagnostic accuracy and increase the risk of missing lesions, especially in preclinical stages. In addition, mammography

analysis is highly dependent on the experience and skills of the radiologist. Lesions such as small microcalcifications, tissue asymmetry, or tumors with blurred margins are easily missed or mistaken for normal structures, especially in atypical cases (Fig. 2).

Related Work

Image classification is one of the most important topics in the computer vision and machine learning community. The goal of an image classification task is to automatically classify an image into one of the predefined classes. The classification of breast cancer subtypes has recently been studied using various methods, in which the application of deep learning techniques in classification, segmentation and early prediction of cancer potential has brought many positive results in early detection and treatment, contributing to reducing mortality rates. Many studies have been conducted to develop techniques for automatic classification and segmentation of tumors on mammograms, in which fuzzy learning techniques and big data processing methods are receiving special attention. Fuzzy learning algorithms such as FCM, EnFCM, and MFCM have proven to be highly effective in medical image processing, especially in the ability to manage uncertainty and ambiguity in data. However, current studies still have many limitations in processing speed and scalability when applied to large amounts of medical image data. Dunn et al. [2] presented a new fuzzy clustering algorithm, developed from ISODATA, to detect "compact and well-separated" (CWS) clusters. The main technique is to construct and optimize a second fuzzy version of the squared error criterion function (J_2), with the solution being the fixed point of the iteration operator. The paper uses 2D synthetic data sets such as three square grids, large and small clusters, two octagonal grids with bridges and halos. The accuracy is not quantified, but evaluated by the membership function value. Through the development process, the FCM algorithm has been widely used in many research works. Szilagyi et al. [3] presented a new algorithm for segmenting brain MRI images using fuzzy techniques. This enhanced version of FCM focuses on significantly reducing computational cost compared to the earlier BCFCM algorithm while maintaining good segmentation

Fig. 2 Shape and density of mass areas in mammogram



quality. The main contribution lies in applying local image filtering and a tuned objective function. However, the study does not address mammography datasets or provide specific accuracy metrics for breast tumor classification. Mohammed et al. [4] applied the FCM technique to MRI images for brain tumor segmentation. Their approach demonstrated high accuracy in delineating tumors and preserved essential image features, aiding the diagnostic process. Han et al. [5] explored the EnFCM algorithm for medical image segmentation. Their method utilizes automatic peak detection to determine the number of classes and peak values, which significantly reduces computational complexity. Testing on 512x512 grayscale images yielded clear segmentation results with reduced processing time. Wang et al. [6] improved the FCM algorithm to enhance segmentation accuracy, particularly for tumors in mammograms. Their method addresses image noise and boosts computational efficiency over traditional FCM approaches. Chen et al. [7] proposed a modified FCM (MFCM) for breast density segmentation in mammograms. Their method involves adaptive local window filtering followed by histogram-based clustering using EnFCM. Testing on noisy synthetic and mammographic images from the EPIC database revealed that MFCM outperforms other algorithms in segmentation accuracy. Wang et al. [8] introduced a multi-scale diffusion filtering method combined with FCM (MsFCM) to improve classification in low-contrast, noisy MRI images. Their approach, tested on the McGill brain MRI database, achieved an overlap ratio of over 90%. Lakshmi et al. [9] employed a modified FCM (M-FCM) algorithm using morphological operations for brain tumor segmentation in MRI images, achieving accuracy up to 98%. Hemanth et al. [10] further refined MFCM by integrating data reduction techniques to lower input size, thereby improving convergence speed while preserving segmentation accuracy. They validated their approach on 256 256 grayscale brain MRI images. FCM algorithms are continuously improved and optimized to achieve high performance in many problems. Touil et al. [11] combined FCM with iterative segmentation for breast segmentation on the MiniMIAS dataset (200 out of 322 images). Despite under-segmentation in low-contrast images, the method achieved a mean clustering measure (CM) of 0.989 and accuracy (ACC) of 0.937. Sadad et al. [12] developed the FCMRG method, integrating FCM with region growing, morphological operations, and hybrid feature extraction (LBP-GLCM, LPQ), followed by classification using Decision Trees or kNN. It achieved 98.2% accuracy on MIAS (109 ROIs) and 91.7% on DDSM (72 ROIs), although it is dependent on preprocessing steps. Chakraborty et al. [13] introduced the Fuzzy Electromagnetism Optimization (FEMO) with Type-2 FCM, demonstrating superior cluster index performance (DB Index, Dunn)

across various medical images (CT, MRI, X-ray), albeit requiring manual cluster selection. Zhang et al. [14] enhanced FCM using Particle Swarm Optimization (PSO) and Mahalanobis distance, achieving 89.33% accuracy on the Iris dataset and 94.94% on the Wine dataset, along with a PSNR of 20.86 on Gaussian noise images. However, performance is sensitive to parameter tuning. Oskoue et al. [15] proposed SSFCM-FWCW, a semi-supervised FCM variant utilizing feature weights and non-Euclidean distances. Tested on 13 datasets, it achieved an average accuracy of 96.66%, improving noise robustness but requiring labeled data and being prone to overfitting. In some recent studies, FCM is integrated into machine learning models and advanced techniques yield significant results in classification problems. Zamri et al. [16] combined FCM with Fuzzy Chebyshev distance and PCA for dimensionality reduction on prostate cancer genomic data (839 samples). The method improved noise immunity and reduced iterations (9 vs. 10), although specific metrics were not reported. Rathinam et al. [17] introduced Adaptive FCM (AFCM) with Opposition-Based Cat Swarm Optimization (OCSO) and VGG-Net, achieving 93.7% accuracy, 90.9% sensitivity, and 95.6% specificity on mammograms. The use of Grey Code Approximation reduced computational complexity. Krishnakumar et al. [18] proposed FCM-DCS, integrating distorted contour features to improve tumor margin detection, achieving 98.76% accuracy on MIAS and 97.9% on DDSM. Thakur et al. [19] developed an Op-CNN model incorporating multi-level semantic segmentation and optimization via Willow Catkin Optimization. Tested on DDSM, it achieved 99.53% accuracy and Dice coefficient of 0.98. Nour et al. [20] combined U-Net with an active contour model (ACM) for Chinese Mammography Database images (5,202 samples), achieving 97.34% accuracy, Dice 0.913, and IoU 0.991. While it shows strong edge refinement and noise resistance, it is computationally intensive. Lozoya et al. (2025) designed a shallow CNN with residual blocks, classifying microcalcifications in INbreast (410 images) and the Mexico set with 99.71% accuracy using only 29,053 parameters. However, it is sensitive to Gaussian noise. Wu et al. [21] proposed a superpixel segmentation model with PRI of 0.92, while Jiang et al. [22] used U-Net for brain tumor segmentation, achieving a Dice score of 0.926. Sun et al. [23] introduced a multi-scale region selection network without requiring ROI annotation, reaching AUC 0.926 on DDSM (3,071 images) and 0.913 on INbreast, though its performance degrades under noise and tumor overlap. Singh et al. [24] combined K-means with region growing and fuzzy enhancement on the MiniMIAS dataset (31 images), achieving 80% accuracy for cancer cases and 87.5% for normal ones. However, results depended heavily on seed selection and dataset size. Berghouse et al. [25] explored attention mechanisms (ECA,

SE, CBAM) combined with CNNs (ResNet50, DenseNet169, RegNetX64), achieving AU-ROC scores above 0.79 on DDSM and INbreast. Although effective in focusing on key regions, the models show performance inconsistency due to varying resolution and risk of overfitting. Loi et al. [26] used radiomic features with LASSO regression to predict breast density, achieving an AUC of 0.76 across 93 patients. Their approach helps reduce radiation exposure through synthetic mammograms.

In our previous study [27], we integrated fuzzy logic techniques with deep learning models such as U-Net and Mask R-CNN to segment tumors in mammographic images. The preprocessing stage focused on preserving critical features of the tumor, including size, shape, and brightness, thereby enhancing the accuracy of localization. Experimental results on the VinDr dataset and clinical data from several hospitals demonstrated the effectiveness of this approach in accurately segmenting tumor regions. In contrast, the current study proposes a fundamentally different approach. Instead of relying on the training of deep learning models which requires extensive computational resources and fully annotated datasets we focus on extracting spatial characteristics and clinically relevant features directly from the input images. These features are preprocessed using the same fuzzy logic methods as in the previous study. However, rather than feeding them into deep learning networks, the segmentation task is performed using a newly proposed method based on an improved version of the FCM algorithm. Specifically, the original FCM algorithm is refined to enhance sensitivity to tumor boundaries and reduce image noise, resulting in more effective segmentation, particularly for tumors with heterogeneous appearances in mammograms. Some of our main contributions in this study included: a) Propose a new method (MoFCM) to improve the FCM algorithm by incorporating fuzzy logic in the preprocessing of the input image to retain important regions for accurate volume segmentation, spatial information, spatial coefficients and standard deviations are calculated appropriately to reduce noise on the input image, b) Construct the MVinDr dataset based on the VinDr dataset in which masks are added based on the opinions of medical experts and hospital doctors to help in accurate volume segmentation, c) Determine the optimal number of clusters for segmentation using the proposed method, d) Improve the clustering time faster than the FCM algorithm, e) Compare and evaluate the accuracy and efficiency of the clustering with previous studies. We conducted experiments on many different scenarios to propose an effective method for tumor segmentation in mammography. Section 3 provides an overview of the algorithms used, Sect. 4 describes the proposed method, Sect. 5 presents the experimental results, and Sect. 6 draws conclusions.

Background

Deep Learning

Computer-Aided Diagnosis (CAD) systems use computers to help medical diagnosticians make judgments. CAD systems are extensively used in medical imaging, such as X-rays, computed tomography (CT), and magnetic resonance imaging (MRI), to help detect abnormalities associated with diseases. citeAbdelrahman2021. Traditional CAD algorithms detect suspicious regions using image processing techniques, which are often rely on a complex set of handcrafted features, including texture, shape, and gray-level intensity that human experts must extract over time. Traditional computer-aided detection can detect breast cancers with excellent accuracy, especially when modern machine learning techniques are used. Traditional CAD systems relying on hand-crafted features, on the other hand, are incapable of handling complicated variations in tumor shape and density distribution, resulting in a high number of false positives. To address the issue of processing hand-crafted features in traditional CAD, some contemporary studies apply advanced Deep Learning models to the stages of CAD systems. Deep learning models may learn and extract high-level features from raw input data, resulting in significantly better classification performance than previous methods. Deep learning, a type of artificial intelligence, has emerged as a valuable tool in medical image segmentation due to its capacity to automatically learn and extract characteristics from complex data. Unlike traditional methods that rely on hand-crafted features, deep learning processes images using deep neural networks, which are especially effective in segmentation challenges like recognizing tumors, organs, or lesions in CT, MRI, or X-ray images. Deep learning network models like InceptionV3 [28], DenseNet, ResNet, VGG19, and Xception are crucial for medical image classification and segmentation as they extract complicated and efficient features. InceptionV3, with its parallel Inception blocks and convolutional decomposition algorithms, improves computational efficiency while achieving high accuracy on classification applications like skin cancer diagnosis. DenseNet121 and DenseNet201 use dense connections to improve information propagation, limit gradient loss, and have been found to be effective in chest X-ray or mammography classification [29]. ResNet152 and ResNet50V2, with skip connections, enable deep network training while preserving accuracy and are widely utilized in MRI or CT analysis [30]. VGG19, with a basic design and a 3x3 kernel, is ideal for transfer learning in skin lesion or X-ray classification [31]. Xception, based on depth-separated convolution, decreases parameters and improves speed. It is utilized in COVID-19 or breast cancer diagnostics [32]. When trained on huge datasets like

ImageNet or specialized medical data, these models perform similarly to or better than human physicians in a variety of tasks. U-Net[33] is a prominent segmentation model with an encoder-decoder structure. It is extremely useful in medical picture segmentation due to its ability to incorporate global and local information. completely Convolutional Networks (FCN), which enable pixel-by-pixel segmentation without the use of a completely connected layer; and advanced models like DeepLab, which use atrous convolution to widen the field of view while preserving high resolution.

Mask R-CNN [34] is an advanced deep learning model developed by the Facebook AI Research (FAIR) team in 2017. It seeks to solve the problem of instance segmentation in computer vision. This model extends Faster R-CNN by including a mask prediction branch for each item, which runs alongside the classification and bounding box regression branches. Mask R-CNN's unique feature is its ability to generate pixel-accurate segmentation masks for each recognized item. This is made feasible by the ROIAlign approach, which improves accuracy when compared to typical ROI Pooling methods. Mask R-CNN has demonstrated outstanding performance on a variety of benchmark datasets, including COCO. It is widely employed in a variety of applications, including autonomous vehicles, medicine, robotics, and document text recognition. Furthermore, this model serves as the foundation for several modern systems, like Detectron2, and continues to play an important role in continuing computer vision research. Furthermore, techniques such as data augmentation, transfer learning, and attention processes are utilized to improve model performance on medical datasets, which are frequently tiny and diverse. These advancements have increased accuracy in applications such as brain, heart, and lung tumor segmentation, allowing doctors to identify and plan treatments more effectively and paving the way for automated systems in medicine.

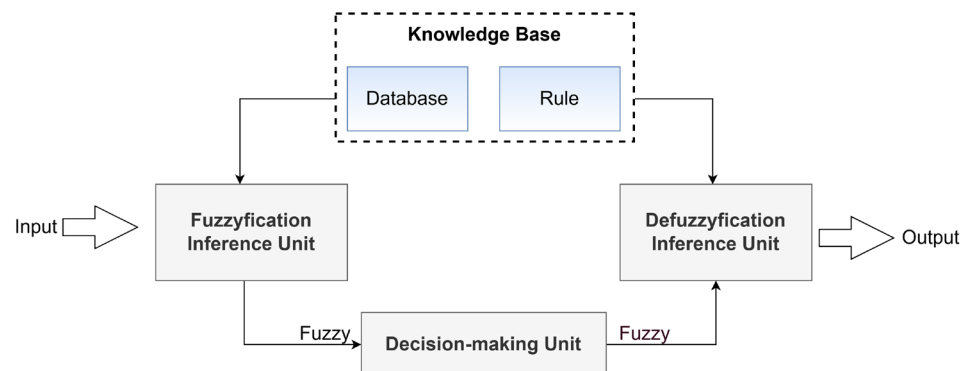
Fuzzy logic

Fuzzy logic [35] is an approach used to handle information and make decisions in situations involving vagueness or uncertainty. In contrast to traditional logic, which operates with strict binary values (0 or 1), fuzzy logic introduces a continuum of values between 0 and 1, allowing it to capture partial truths. Fuzzy machine learning integrates these fuzzy principles into machine learning algorithms [36], making them particularly suitable for analyzing uncertain or imprecise data, such as that found in medical imaging. These models help interpret and identify abnormal image regions more effectively, enhancing diagnostic accuracy and classification performance. A fuzzy inference system (refer to Fig. 3) facilitates decision-making through a structured set of fuzzy rules and a comprehensive knowledge base.

Fuzzy Clustering

Clustering is a frequently used technique in unsupervised learning that groups data points based on their commonalities. This approach has many practical applications in a variety of fields. The core premise of clustering is to divide a set of data into subgroups or clusters, ensuring that observations within the same cluster share a similar level of similarity. Fuzzy clustering, also known as fuzzy clustering, is a form of clustering algorithm in machine learning that allows each data point to belong to numerous clusters with variable degrees of similarity, as opposed to hard clustering, which assigns each point to a single cluster. Unlike standard methods such as K-means or hierarchical clustering, this method allows each data point to belong to numerous clusters with various membership degrees (values ranging from 0 to 1). Clustering seeks to organize data points so that they are as similar as possible inside the same cluster, but as dissimilar as possible between clusters. FCM is a well-known fuzzy clustering technique developed by Dunn [2] and further enhanced by Bezdek [38]. FCM operates by assigning a degree of membership to each cluster based on the distance between the data point and the cluster center. The algorithm

Fig. 3 Fuzzy inference system (FIS) [37]



minimizes an objective function where data points closer to the cluster centers have higher membership values. A significant parameter in FCM is m (with $m > 1$), which controls the degree of "fuzziness" in the clustering process; larger values of m result in more fuzziness. FCM has numerous applications in domains such as image processing, pattern identification, medicine, and data mining, especially when data overlaps or cluster boundaries are unclear. Unlike the K-Means method, FCM provides more detailed insights into how each data point links to the clusters, which improves understanding of data structure. Depending on the data and application, measures like distance, connection, or intensity are used to assess similarity. The formula symbols presented in the paper are listed in the Table 1.

Fuzzy C-means

The FCM algorithm is a widely used fuzzy clustering method. It assigns each data point to multiple clusters with different membership degrees by minimizing an objective function that balances the distance from data points to cluster centers and the degree of fuzziness. Given an input image I with N pixels. The objective function is defined as follows:

$$J_{FCM}(t, v) = \sum_{j=1}^N \sum_{i=1}^Z t_{ij}^m d^2(x_j, v_i) \quad (1)$$

The initial cluster center is calculated based on the formula:

$$Z_k = \frac{1}{1 + N_k} \sum_{x_j \in N_k} x_j \quad (2)$$

where x_j is the intensity of neighboring pixels of pixel x_k , and N_k is the number of pixels in the neighborhood. The membership matrix, denoted t , is an $N \times k$ matrix, where N is the number of data points and k is the number of clusters. For each $i=1, \dots, Z$ and $j=1, \dots, N$ the element t_{ij} represents the membership degree of data point x_i in cluster j , $t_{ij} = t_j(x_i)$. The membership function t_{ij} is defined as follows:

$$t_{ij} = \frac{1}{\sum_{k=1}^Z \left(\frac{\|x_j - z_i\|^2}{\|x_j - z_k\|^2} \right)^{\frac{2}{m-1}}} \quad (3)$$

Where: i is the index of the data point, x_j is the j th data point, Z is the total number of clusters, m is the fuzzification parameter in this case $m=2$ which determines the fuzzification level of the clusters. $|x_j - z_i|$ is the distance between the data point x_j and the cluster center z_i . $|x_j - z_k|$ is the euclidean distance between the data point x_j and the cluster center z_k . The function updates the cluster centers of all data points x_i with the weight being the membership matrix t_{ij} and the fuzzification value m used to adjust the influence level of t_{ij} .

$$z_i = \frac{\sum_{j=1}^q (t_{ij})^m x_j}{\sum_{j=1}^q (t_{ij})^m} \quad (4)$$

-
- 1: **Input:** Original image with N pixels I
 - 2: Initialization
 C : Number of clusters
 m : Fuzzy coefficient ($m = 2$)
 ε : Stop threshold
 - 3: Randomly initialize the centers of clusters z_i , $i = 1, \dots, Z$
 - 4: **repeat**
 - 5: Compute membership t_{ij} using cluster center z_i by Eq.3
 - 6: Update cluster center z_i using t_{ij} by Eq.4
 - 7: **until** $\|z^{(t+1)} - z^{(t)}\| < \varepsilon$
 - 8: **for** $i = 1, \dots, Z$ **do**
 - 9: Create the i segmented image
 - 10: **end for**
 - 11: Returns the image segmented into Z regions.
-

Algorithm 1 FCM algorithm

Enhanced Fuzzy C-means (EnFCM)

EnFCM is an improved version of FCM, designed to overcome some limitations of FCM, such as sensitivity to noise, dependence on cluster center initialization, or performance

on complex data. In this paper, we use noise filtering techniques to reduce the impact of outlier data points. The objective function of the algorithm is presented in Eq. 5.

$$J_{\text{EnFCM}}(t, z) = \sum_{j=1}^N \sum_{i=1}^Z t_{ij}^m \|x_j - z_i\|^2 + \alpha \sum_{j=1}^N \sum_{i=1}^Z t_{ij}^m \|x_j - \hat{x}_i\|^2 \quad (5)$$

When performing the algorithm on image data, the algorithm applies a local filtering to each pixel, based on the described neighborhood. Where the adaptive local filter value $\hat{X}_k$ is given by the Eq. 6:

$$\hat{X}_k = \frac{\sum_{r \in N_k} Z_{kr} \cdot x_r}{\sum_{r \in N_k} Z_{kr}} \quad (6)$$

The local filter can be tuned through the coefficient α and the neighborhood size.

-
- 1: **Input:** Original image with N pixels I
 - 2: Apply local filter
 - 3: Calculate number of different intensity levels
 - 4: Initialize
 Z : Number of clusters
 m : Fuzzy coefficient
 ε : Stop threshold
 - 5: Initialize specific initial cluster using Eq.6
 - 6: **repeat**
 - 7: Calculate membership matrix t_{ij} using cluster center z_i by Eq.3
 - 8: Update cluster center z_i using t_{ij} by Eq.4
 - 9: **until** $|z^{(t+1)} - z^{(t)}| < \varepsilon$
 - 10: **for** $i = 1, \dots, Z$ **do**
 - 11: Create the i segmented image
 - 12: **end for**
 - 13: Returns the image segmented into Z regions.
-

Algorithm 2 EnFCM algorithm

Evaluation Metrics

Model evaluation is the process of evaluating the performance and accuracy of proposed algorithms. Model evaluation helps to evaluate the ability of the model to solve a particular problem and can be used to compare different models. In clustering problems, the choice of the number of clusters is an important issue to help the model achieve good performance while saving computational costs.

Segmentation accuracy: measures the percentage of correctly predicted pixels compared to the total number of pixels in the image.

$$Seg_acc = \frac{TruePos. + TrueNeg.}{N} \times 100 \quad (7)$$

where *TruePos.* (True Positive) is the number of pixels segmented as mass region and *TrueNeg.* (True Negative) is the number of pixels that are not mass region out of the total number of pixels N .

Xie-Beni Index (XB): The XB index measures the separation between clusters and the compactness within clusters. The lower the XB index, the better the clustering. For each cluster center z_i of the membership matrix t_{ij} , we have the following formula to calculate XB:

$$Xie_Beni = \frac{\sum_{i=1}^Z \sum_{j=1}^N t_{ij}^m \|x_j - z_i\|^2}{N \cdot \min_{i \neq k} \|x_j - z_i\|^2} \quad (8)$$

Where: $\|x_j - z_i\|^2$ The squared Euclidean distance between the data point x_j and the cluster center z_i . $\min_{i \neq k} \|x_j - z_i\|^2$ the least squared distance between any two cluster centers.

Where m is the fuzzy coefficient, N is the total number of clustered pixels. The numerator measures the cluster closeness by calculating the sum of the squared weights from the data point to the cluster center, the closer the point is to the cluster center, the smaller this index is. The denominator measures the degree of separation by calculating the smallest distance between cluster centers, the larger the value, the more separated the clusters are.

Partition Coefficient (PC): The PC index is used to measure how clearly and reliably data points belong to clusters. A higher PC index shows the effectiveness of clustering. This index is used to assess the reliability of clustering and is used to compare clustering results with different numbers of clusters given by the formula:

$$Part_coef = \frac{1}{n} \sum_{i=1}^Z \sum_{j=1}^N t_{ij}^2 \quad (9)$$

Table 1 Common symbols used in the paper

Symbol	Description
x_j	Intensity value of j^{th} pixel of data point
z_i	Center of i^{th} cluster
t_{ij}	Membership matrix
Z	Total number of clusters
N	Total number of pixels in image
ε	Convergence threshold
m	Fuzzy parameter
q	Number of distinct intensity levels in filtered image
σ_k	Standard deviation of intensity in neighborhood
Z_{kr}	Spatial deviation between pixel r and neighborhood k
$I^{filtered}$	Image after applying filter
FCM	Fuzzy C-Means
EnFCM	Enhanced Fuzzy C-Means
PC	Partition Coefficient
XB	Xie-Beni Index
AFCM	Adaptive Fuzzy C-Means
SSFCM-FWCW	Semi-Supervised Fuzzy C-Means with Feature Weights and Cluster Weights

Where: N is the number of data points, Z is the number of clusters, t_{ij} is the degree of belonging of data point x_j to cluster i . The PC value lies in the range $[1/Z, 1]$ representing the degree to which a data point is completely or equally dependent on a cluster.

Methodology

FCM and EnFCM algorithms are designed to address challenges in medical image processing, such as noise, low contrast, and overlapping structures. Unlike traditional image

processing filters such as Box filters, Gaussian filters, or bilateral filters, which mainly focus on smoothing or denoising while preserving edges, FCM and EnFCM combine spatial information and fuzzy logic to enhance segmentation accuracy. A simple box filter reduces noise but blurs important details, making it unsuitable for distinguishing subtle features such as microcalcifications in mammography. A Gaussian filter smooths images effectively but may blur the boundary between lesions and non-lesions. A bilateral filter preserves edges better than a Gaussian filter but struggles with high noise and complex overlapping structures, which are common in mammography. Therefore, the solution integrates a set of fuzzy rules based on tumor features (e.g., shape, brightness, and density) and spatial constraints to improve cluster strength and segmentation performance, which provides promising performance. MoFCM enhances traditional FCM by incorporating local filtering, which reduces noise while retaining essential image details. These improvements make MoFCM particularly effective in tumor segmentation in mammography, where accurate differentiation of tissue types is critical. Compared with methods such as FCM with spatial information (FCM-S) and MFCM that only focus on improving the objective function or applying local filters, MoFCM stands out by integrating a fuzzy inference engine based on clinically validated attributes (size, shape, brightness) in the preprocessing step something that methods such as AFCM or SSFCM-FWCW do not do. In addition, MoFCM keeps the model simple, does not require labeling like SSFCM or complex optimization like FEMO, making it easy to apply in practice. Developed from the basic FCM algorithm, the integration of spatial uncertainty X_k and spatial coefficient Z_{kr} helps the algorithm improve the clustering ability in many cases with noise on the input data. The standard deviation of the neighborhood

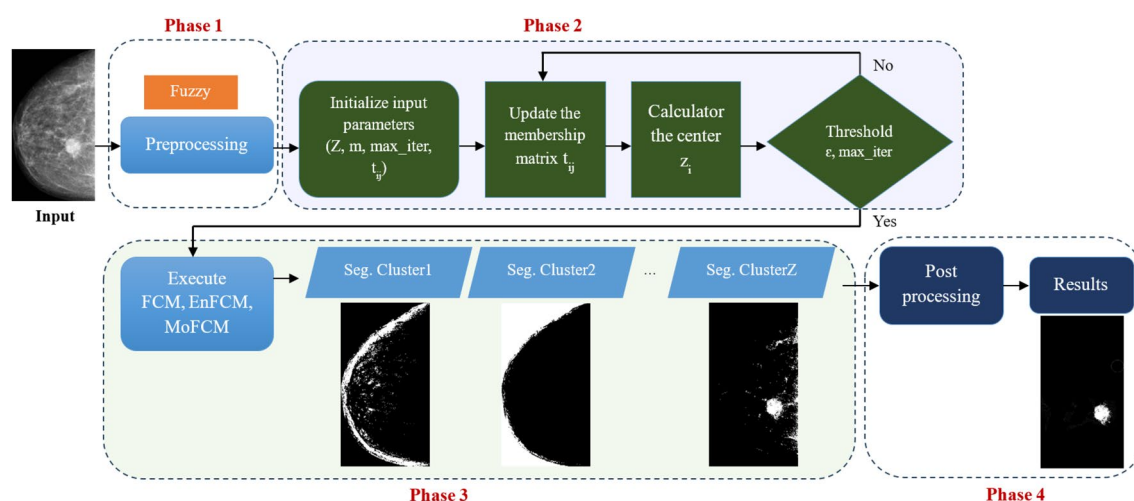
**Fig. 4** Proposed model for mass segmentation on mammograms

Fig. 5 The preprocessing of the input image applies fuzzy logic

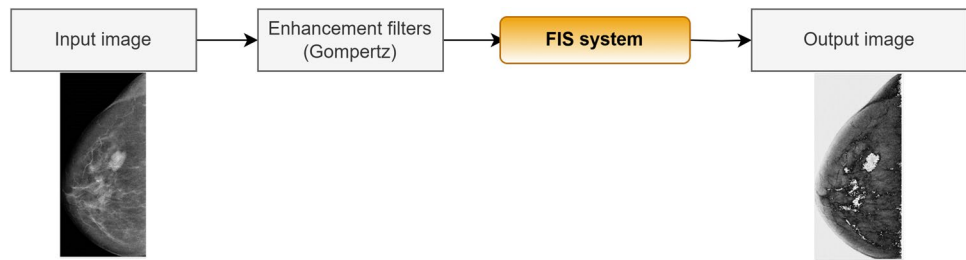


Table 2 Fuzzy logic rule set in image preprocessing

Rules	Size	Shape	Brightness	Enhancement
Rule 1	Large	Round, oval	High	High
Rule 2	Medium	Round, oval	Medium	Medium
Rule 3	Small	–	Low	Low

N_k measures the dispersion of pixel intensities in a window k with a window size of 5×5 and is presented as follows:

$$\sigma_k = \sqrt{\frac{1}{N_k} \sum_{r \in N_k} (X_r - \hat{X}_k)^2} \quad (10)$$

The spatial weighting factor Z_{kr} is based on the exponential function of the distance d_{kr} which is the euclidean distance between pixel k and r in space.

$$Z_{kr} = e^{-d_{kr}} \text{ if } r \in N_k \quad (11)$$

Figure 4 illustrates the overall pipeline of the proposed MoFCM algorithm, showing the four major stages: preprocessing with fuzzy logic, cluster center initialization and updating, segmentation using clustering, and final post-processing. This architecture highlights how MoFCM integrates domain knowledge (via fuzzy rules) and spatial filtering to improve segmentation accuracy. The implementation details are divided into 4 phases as follows:

Phase 1 (Preprocessing using fuzzy logic): The fuzzy block (Fig. 5) applies fuzzy logic techniques to enhance tumor-containing regions, reducing noise and improving contrast. This step distinguishes MoFCM from traditional FCM. Unlike traditional filters such as Gaussian, bilateral or box filters which process the whole image uniformly, the MoFCM approach integrates spatial context and fuzzy logic rules to selectively enhance tumor-like regions. While Gaussian and bilateral filters are effective for smoothing and edge preservation, they do not consider semantic tumor features (shape, brightness, size). In contrast, our fuzzy preprocessing stage allows selective enhancement based on medical knowledge and tumor characteristics verified in prior studies.

The process of image preprocessing using fuzzy logic techniques can be outlined as follows. First, preprocessing and enhanced filtering are applied Gompertz function.

Next, a fuzzy block is implemented, which involves a fuzzy inference system to assess the level of mass enhancement in medical images based on three key attributes size, shape, and brightness. The fuzzy member functions define the levels of these attributes, where size is categorized as small, medium, or large, shape as round, irregular, or oval, and brightness as low, medium, or high. Fuzzy rules are then applied to generate output images that preserve mass-containing regions, retaining the pixels within these areas. The set of rules is presented in the Table 2:

The result of the preprocessing process is to retain the features that contain mass on the input image to conduct clustering in phase 2.

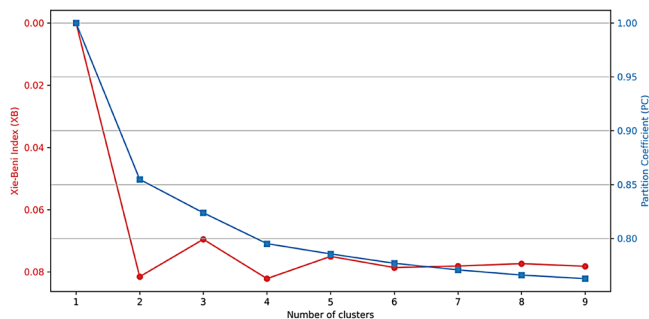
Phase 2 (Compute and update cluster centers): The input data is initialized with input parameters including: the number of clusters z_i ($i=1, \dots, Z$), the fuzzy coefficient $m=2$, the maximum number of iterations $\text{max_iter}=100$, and the initial membership matrix t_{ij} . Next, the algorithm updates the membership matrix t_{ij} based on the distance from each pixel to the cluster centers. Then, the cluster centers z_i are recalculated. After each update, the algorithm checks the convergence condition based on the error threshold ϵ or when the maximum number of iterations max_iter is reached.

Phase 3 (Execute algorithms): The three algorithms FCM, EnFCM, and MoFCM are executed independently, each with its specific preprocessing and filtering steps, as illustrated in Fig. 4. The proposed algorithm (MoFCM) integrates fuzzy preprocessing and spatial-aware filtering, improving upon FCM and EnFCM in accuracy and robustness.

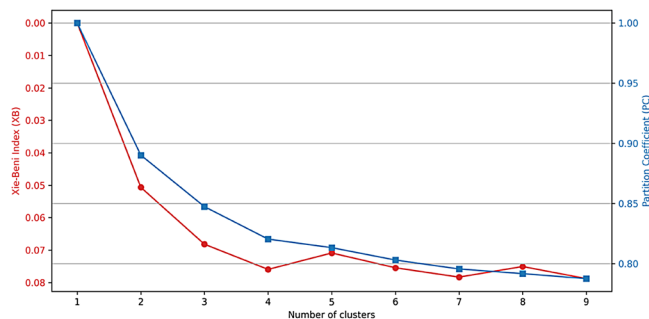
Phase 4 (Post-processing the output on the cluster): After being clustered on clusters, the image undergoes post-processing based on mass characteristics such as size, shape, and density. On the clustered image result from z_6 , small bright regions are removed based on a predefined threshold. Finally, a mapping technique is used to convert the binary image into a grayscale image.

Table 3 XB and PC values correspond to the number of clusters on the DDSM, MViDr and Our datasets

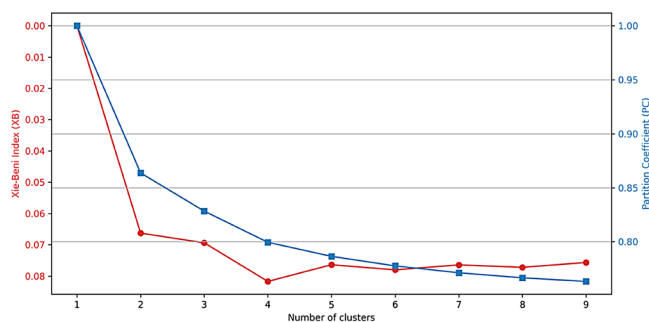
Number of clusters	DDSM		MViDr		Our dataset	
	XB	PC	XB	PC	XB	PC
1	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000
2	0.0815	0.8548	0.0507	0.8902	0.0663	0.8637
3	0.0695	0.8237	0.0682	0.8474	0.0694	0.8283
4	0.0822	0.7951	0.0760	0.8204	0.0817	0.7995
5	0.0750	0.7855	0.0709	0.8133	0.0764	0.7864
6	0.0786	0.7769	0.0754	0.8030	0.0780	0.7776
7	0.0781	0.7708	0.0783	0.7956	0.0764	0.7712
8	0.0774	0.7659	0.0751	0.7918	0.0772	0.7666
9	0.0782	0.7626	0.0788	0.7876	0.0757	0.7632



(a) DDSM dataset



(b) MViDr dataset



(c) Our dataset

Fig. 6 Average XB and PC by number of clusters of 3 datasets

```

1: Input: Original image  $I$  with  $N$  pixels
2:   Number of clusters  $Z$ , Fuzzy coefficient  $m$ , Convergence threshold  $\varepsilon$ 
3: Step 1: Preprocessing
4:   Apply fuzzy block (Fig.5) on image  $I$ 
5: Step 2: Initialization
6:   Randomly initialize cluster centers  $z_i, i = 1, \dots, Z$ 
7: for each  $5 \times 5$  window centered at pixel  $k$  do
8:   Compute the standard deviation  $\sigma_k$  of intensity values in the window using
   Eq.10
9:   Identify the set of pixels  $r \in N_k$  where  $|x_r - \hat{x}_k| \leq \sigma_k$ 
10:  Compute the spatial deviation  $Z_{kr}$  for each pixel  $r$  using Eq.11
11:  Apply a local filter to create the filtered image  $I^{filtered}$ 
12:  Compute the intensity histogram of the filtered image
13: end for
14: Step 3: Update Membership Matrix
15: repeat
16:   Compute the membership matrix  $t_{ij}$  using Eq.3
17:   Update cluster centers  $z_i$  using Eq.4
18: until Convergence criterion is met
19: Step 4: Clustering
20: for each pixel  $k$  do
21:   Assign pixel  $k$  to the cluster  $i$  with the highest membership value
22:   Construct the segmented image  $I$ 
23: end for
24: Output: Segmented image  $I$  with  $Z$  regions

```

Algorithm 3 Proposed Method (MoFCM algorithm)

Experimental Results

Datasets and Installation Environment

In this study, we focus on mammogram images containing masses from three datasets to evaluate the performance of the proposed algorithms. The first dataset, DDSM [39], consists of 10,239 images, including 912 benign and 784 malignant cases, collected from various medical centers in the United States. It provides comprehensive clinical and histological annotations. To assess the model's effectiveness in a domestic context, we also utilize the VinDr-Mammo [40] dataset, introduced in 2022, which includes 20,000 images from 5,000 cases captured from four standard mammographic views (CC and MLO). This dataset contains Birads 1-5 classifications and bounding box annotations for lesion regions. All datasets used in this research are either

publicly available or collected with proper ethical approval. Specifically, DDSM and VinDr-Mammo are open-access datasets widely used in the medical imaging community. To enhance evaluation, we created an extended version called MVinDr, derived from VinDr-Mammo, by adding expert-annotated segmentation masks based on consultations with radiologists. This dataset comprises 1,226 masked images and adheres to the data usage policies of the original source. In addition, we collected a private dataset from local hospitals, consisting of 1,045 images, which were classified and annotated by medical experts. This dataset was anonymized and gathered under the approval of local medical ethics committees, ensuring compliance with all applicable regulations. The system operates within a Python environment on Kaggle, powered by an Intel Xeon 2.00 GHz processor and 30GB of RAM (Table 3).

Table 4 Proposed experimental scenarios

Scenario	Algorithm	Fuzzy block	Z	m	ϵ	Iteration
1	FCM	—	6	2	0.05	100
2	EnFCM	—	6	2	0.05	100
3	MoFCM	—	6	2	0.05	100
4	FCM	preprocessing	6	2	0.05	100
5	EnFCM	preprocessing	6	2	0.05	100
6	MoFCM	preprocessing	6	2	0.05	100

Dataset	Input	Ground truth	z1	z2	z3	z4	z5	z6
DDSM								
MViDr								
Our dataset								

Fig. 7 Clustering results on FCM algorithm

Results of Evaluating the Number of Clusters

In order to determine the appropriate number of clusters on the 3 datasets DDSM, MViDr, our dataset, we evaluated the clustering level with 2 indexes XB according to the formula Eq. 8 and PC according to the formula Eq. 9. The results of the clustering level evaluation are presented in the chart and table (Fig. 6).

Accordingly, we choose the corresponding XB with the smallest index and PC with the highest score. Because the PC index will decrease as the number of clusters increases, in this case the XB index will be used to determine the number of main clusters, in which the PC index will be used to support appropriate decision making. When combining both values on the same dataset, it helps determine the appropriate number of clusters on the 3 datasets. The results show

that the number of clusters = 6 will be appropriate to perform the evaluation on all 3 datasets.

Applicable Scenarios

To evaluate the effectiveness of the proposed method and compare it with our group's previous studies, we propose nine experimental scenarios. These scenarios build upon our prior research. During the experiments, we utilize the model evaluation metrics outlined in Section 3, along with steps to analyze the achieved efficiency, to aid in selecting the most suitable model for mammography segmentation. The details of the proposed scenarios are presented in the Table 4.

In the paper, scenarios are set up to evaluate the performance of FCM, EnFCM and MoFCM algorithms with fixed parameters: $Z = 6$, $m = 2$, $\epsilon = 0.05$ and the number of iterations is 100. Specifically, in the first 3 scenarios we

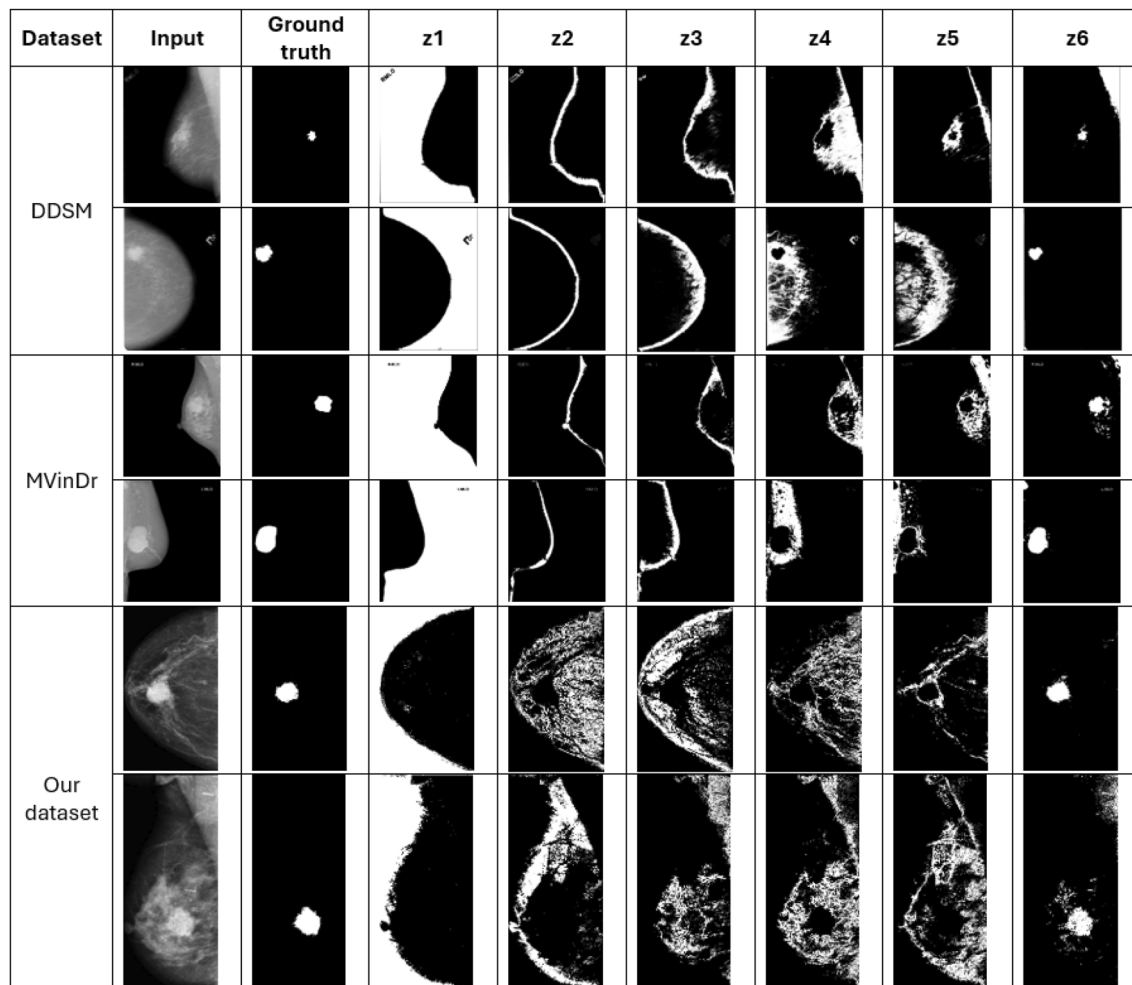


Fig. 8 Clustering results on EnFCM algorithm

use the default input values to evaluate the proposed model. Then, in scenarios 4, 5 and 6, FCM, EnFCM and MoFCM are applied respectively with input images that are applied with fuzzy blocks in image enhancement, to compare the efficiency of algorithms and the efficiency of image preprocessing using fuzzy under different conditions on the same parameter set. Based on the proposed scenarios, we conduct experiments to evaluate the effectiveness of the proposed method, the results are presented in the next section.

Clustering Results and Algorithm Evaluation

Figs. 7, 8, 9 present visual comparisons of the segmentation outputs from the FCM, EnFCM, and MoFCM algorithms, respectively. Each figure includes clustered regions (z1–z6) extracted from the DDSM, MViDr, and our private dataset. These comparisons visually demonstrate the progression in segmentation quality achieved by the proposed approach.

Based on the image of the clusters from the FCM algorithm in Fig. 7, which is applicable to three sets of DDSM

data, MViDr, and separate data sets, we can observe a clear distinction between the z1 to z6 clusters. The z1 cluster distinctly illustrates the outer border of the breast tissue, whereas the z2 cluster primarily emphasizes the base and breast tissue. The z3 and z4 clusters demonstrate the separation between abnormal and normal tissue areas, particularly evident in the MViDr data set and its associated data kit. The z5 cluster exhibits more noise due to the complex structures present, especially in the DDSM data and separate data. Finally, the z6 cluster indicates that the suspected lesions are relatively clearly delineated across all three data sets; however, the MViDr data set features sharper clusters than the DDSM and its own dataset, highlighting differences in image quality and noise levels (Fig. 8).

The EnFCM algorithm shows its clustering results with groups z1 to z6, which illustrate the shift from the outer edge to the areas with lesions in the breast image. On the DDSM and MViDr datasets, the clustering is clear and consistent. Specifically, cluster z1 helps to separate the outer edge of the breast tissue, cluster z2 highlights the mammary gland

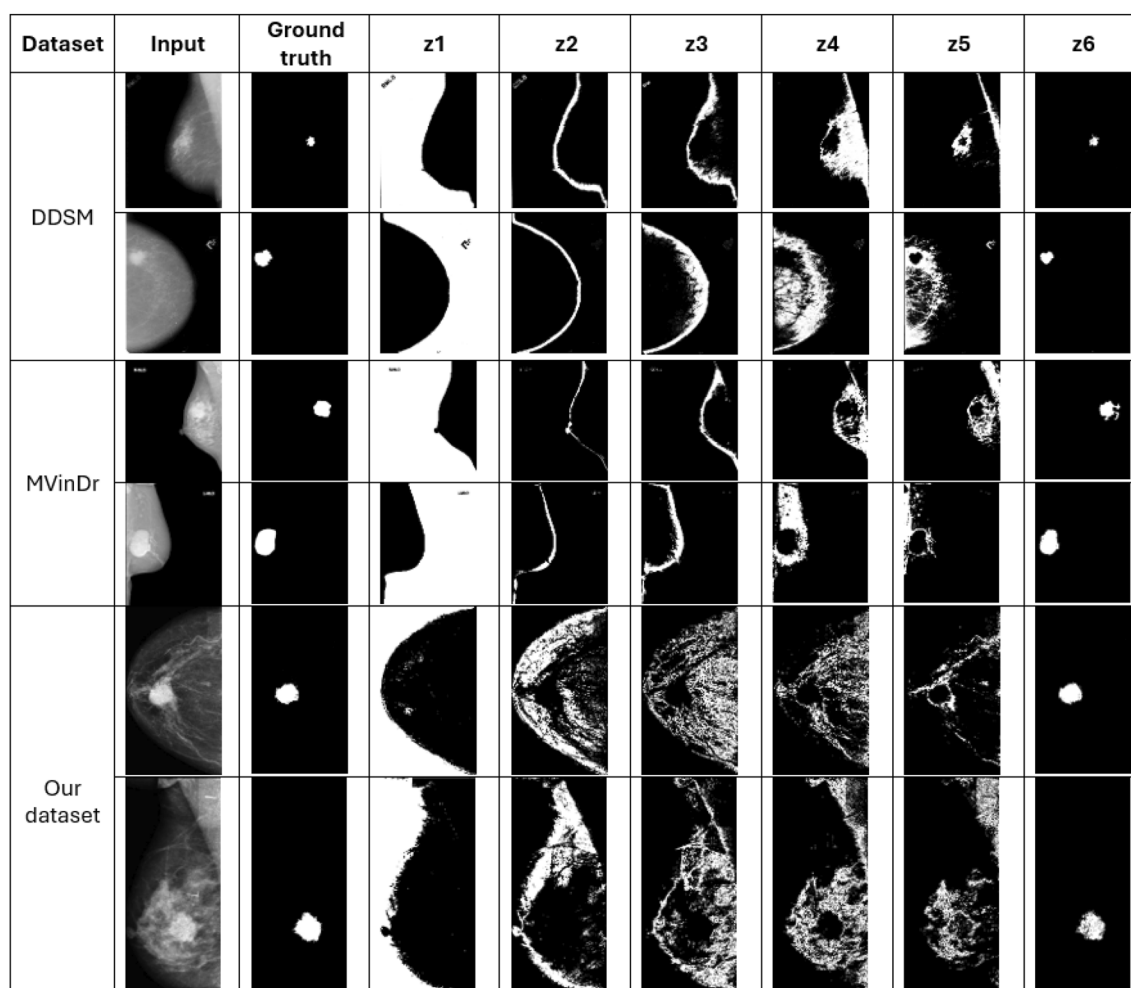


Fig. 9 Clustering results on MoFCM algorithm

Table 5 Table comparing the running times of the three algorithms on the three datasets

Algor. \ Datasets	DDSM (seconds)	MViDr (seconds)	Our dataset (seconds)
FCM	12761	11769	714
EnFCM	5118	9078	610
MoFCM	12383	10534	620

Table 6 Comparison table of partition accuracy of 3 algorithms respectively on 3 datasets

Scenario	DDSM	MViDr	Our dataset
1	0.862	0.897	0.908
2	0.875	0.886	0.929
3	0.882	0.896	0.930
4	0.902	0.909	0.931
5	0.904	0.915	0.935
6	0.914	0.929	0.947

and soft tissue structures, clusters z3 and z4 continue to separate the soft tissue and mammary gland regions in more detail. Cluster z5 begins to emphasize the lesion or tumor regions, and cluster z6 focuses on defining these regions more clearly. On our dataset, the clustering has significant noise, especially in clusters z3 to z6. However, cluster z6 still accurately identifies the mass region in its output.

The clustering process of the proposed method is shown in Fig. 9. The clustering results of z1 to z6 show the transition from the outer edge to the lesion regions inside the mammogram. On the DDSM and MViDr datasets, the clustering is clear and stable. Specifically, cluster z1 clearly identifies the outer edge of the breast tissue, clusters z2 and z3 highlight the mammary gland and soft tissue structures in detail. In clusters z4 and z5, the lesion regions are clearly highlighted with less noise, especially on the MViDr dataset. Cluster z6 shows the ability to separate the lesion regions clearly and consistently. On our dataset, although the input data has noise, the output of the algorithm is still segmented accurately. Based on the clustering results of three algorithms

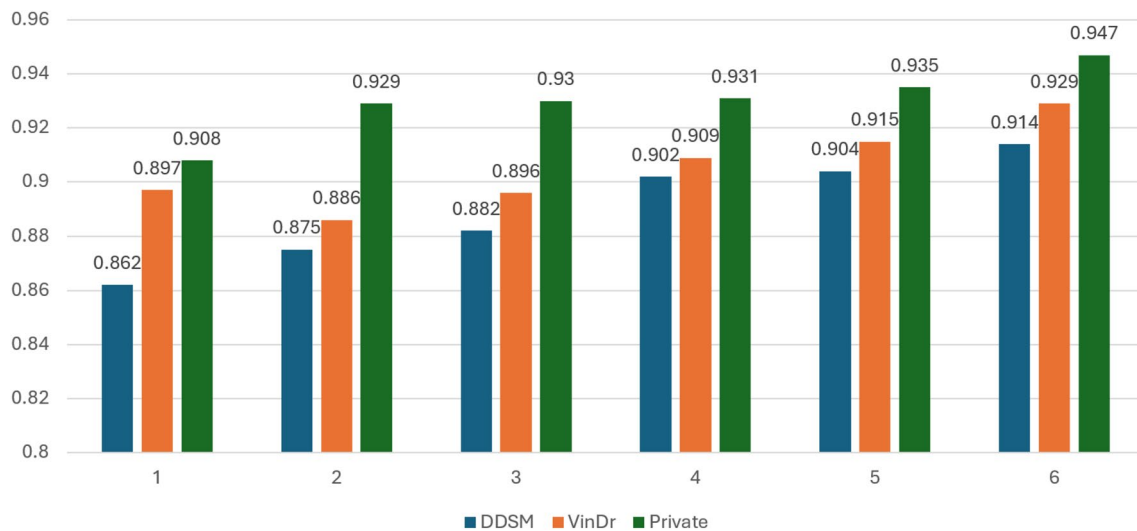


Fig. 10 Accuracy comparison chart between scenarios

Table 7 Comparison of accuracy with previous research works

Authors	Techniques	Dataset	Accuracy(%)
Singh et al.[24]	K-means, region growing	MiniMIAS	87.5
Sadad et al.[12]	FCMRG, hybrid feature extraction	MIAS, DDSM	91.7
Wu et al.[21]	Superpixel segmentation	-	92.0
Sun et al.[23]	Multi-scale area selection network	CBIS-DDSM, INbreast	92.6
Rathinam et al.[17]	GCAP, AFCM, OCSO, DCNN (VGG-Net)	Mammogram	93.7
Nour et al.[20]	U-Net, ACM, preprocessing, data augmentation	Chinese Mammography	91.3
Previous work[27]	Mask R-CNN, U-Net	Our dataset	92.5
Proposed method	FCM, EnFCM	DDSM, MVinDr, our dataset	92.9
Proposed method	MoFCM	DDSM, MVinDr, our dataset	94.7

with three different evaluation datasets, the difference in clustering process is shown. On the FCM algorithm, the segmentation results show that the boundaries between tissue regions are unclear and many regions are mixed in color, making it difficult to distinguish damaged regions from normal tissue. The EnFCM algorithm improves over FCM when the segmented regions become clearer, especially the regions suspected of being damaged. The EnFCM algorithm improves on the output results but still has noise and loss of detail in some border regions. With the MoFCM method, the segmentation quality is improved through the application of fuzzy logic to help distinguish more clearly between regions, especially regions containing mass. In addition to achieving clear separation between bright and dark regions,

MoFCM significantly reduces noise in the z6 cluster. The overall output for each cluster is presented as follows:

- z1 tends to focus on the outer edge of the breast tissue, with strong light intensity, easily distinguishable from the black background.
- z2 tends to focus on dark background or tissue unrelated to the lesion.
- z3 shows the glandular tissue structure more clearly, but there is a lot of noise present in the entire image, reducing the focus on the lesion.
- z4 and z5 begin to show fine and uneven tissue details within the breast tissue, the noise is still significant, making it difficult to isolate the lesion.
- z6 is a cluster that clearly shows the mass region; the bright areas mostly match the actual mask. This is also the base cluster for the final synthesis result.

In the results column, the final segmentation results show that the lesion regions are relatively well separated from the original mask, although there are still some small noise points around. Overall, the segmentation process through cluster extraction improves the ability to isolate lesions from the complex tissue background of mammograms. In addition to segmentation, we compare the running times of the three algorithms present them in Table 5.

Based on the execution time table of the three algorithms FCM, EnFCM and MoFCM on the three datasets DDSM, MVinDr and our dataset, it can be seen that EnFCM has the best performance with the lowest time on all three datasets: 5118 s (DDSM), 9078 s (MVinDr) and 610 s (Our dataset). The FCM algorithm has a slower execution time than the other two algorithms. Our proposed algorithm has a slower time than EnFCM but still has a better improvement than

the FCM algorithm. Next, we evaluate the accuracy of all three algorithms, the results are presented in Table 6.

Based on Table 6 on the segmentation accuracy of the 3 algorithms on the 3 datasets DDSM, MViDr and our dataset, we can clearly see the difference between the scenarios without fuzzy (1, 2, 3) and with fuzzy (4, 5, 6). Specifically, in the scenarios without fuzzy, the accuracy on the DDSM set ranges from 0.862 (scenario 1) to 0.882 (Sce. 3), on MViDr from 0.886 (Sce. 2) to 0.896 (Sce. 3), and on our dataset from 0.908 (Sce. 1) to 0.930 (Sce. 3). Meanwhile, the fuzzy scenarios show significantly higher accuracy: on DDSM from 0.902 (Sce. 4) to 0.914 (Sce. 6), on MViDr from 0.909 (Sce. 4) to 0.929 (Sce. 6), and on our dataset from 0.931 (Sce. 4) to 0.949 (Sce. 6). Overall, the fuzzy scenarios (4, 5, 6) outperform the non-fuzzy scenarios (1, 2, 3) on all three datasets, especially on our dataset set with the highest accuracy of 0.949 in scenario 6, demonstrating that applying fuzzy significantly improves the partitioning performance of the algorithms (Fig. 10).

The comparison chart of the segmentation accuracy of the six scenarios on the three datasets DDSM, MViDr and our dataset shows that the overall performance of the scenarios tends to increase from scenario 1 to scenario 6. Specifically, our dataset set (green) always achieves the highest accuracy in all scenarios, ranging from about 0.91 in scenario 1 to nearly 0.95 in scenario 6, indicating that this dataset is easier to segment. Meanwhile, the DDSM set (blue) has the lowest accuracy, starting from 0.86 in scenario 1 and gradually increasing to 0.91 in scenario 6. The MViDr set (orange) is in the middle, with accuracy ranging from 0.89 to 0.93 across scenarios. Overall, scenario 6 achieved the highest performance on all three datasets, demonstrating the effectiveness of the MoFCM algorithm after the improvement process and demonstrating the segmentation performance on the DDSM set. On the other hand, the use of Fuzzy preprocessing of the input image to highlight the tumor-containing regions significantly improved the segmentation ability on more complex datasets (Table 7).

Although deep learning models such as U-Net and Mask R-CNN have been proven to be highly effective in medical image segmentation, training and optimizing these models requires large resources, detailed labeled data, and long training times. In this study, the goal is to build a lightweight solution that is feasible in resource-constrained environments such as provincial hospitals. Therefore, MoFCM is proposed as an easy-to-deploy model that does not require deep learning but still achieves 94.7% accuracy, which is higher than the 92.5% result in the group's previous deep learning study [27].

Conclusion

This study presents an effective approach for medical image segmentation, focusing on accurate mass delineation in mammograms to diagnose breast cancer, helping to detect breast cancer risks early. While deep learning methods remain state-of-the-art for segmentation, MoFCM offers a compelling alternative for low-resource contexts by delivering competitive accuracy without the need for training, large labeled datasets, or high computational overhead. This makes MoFCM more accessible and easier to deploy in real-world clinical environments. Our study introduces a new method MoFCM, which improves the traditional FCM algorithm through the image preprocessing step by constructing a fuzzy rule set based on important tumor features that have been verified in previous studies and integrating spatial information to minimize the impact of noise, improve cluster strength, and provide superior segmentation performance, helping to accurately segment masses on mammograms. Determining the number of clusters is an important issue to evaluate the efficiency and quality of clustering and comparing the evaluation on algorithms, the XB index and PC, helping to give a suitable number of clusters of 6 for effective segmentation on DDSM, MViDr and our datasets. In previous research, most of the studies performed the evaluation based on bounding box on VinDr dataset, to extend and support accurate segmentation, we constructed MvinDr dataset with masks segmented by experts as a basis for comparison with public datasets. To evaluate the effectiveness of the proposed method, we performed comparisons on multiple scenarios and on clustering evaluation metrics. Experimental results demonstrate that MoFCM outperforms both FCM and EnFCM, achieving the highest segmentation accuracy of 94.7% on our dataset, with 92.9% on MViDr and 91.4% on DDSM. The application of fuzzy preprocessing has significantly improved the algorithm's ability to segment volume regions, especially in complex datasets with noise. The proposed method also improves clustering time better than FCM algorithm.

Besides, the proposed method still has certain limitations. The computation time of the three algorithms is still high, especially on the DDSM and MViDr dataset. This will limit some real-time applications. The performance of the method depends on the quality of the input images, especially the noise phenomenon in low-quality images affects the clustering process. Future research will focus on improving the MoFCM algorithm to reduce the computational complexity, possibly through parallel computing or large data processing environments. We will also combine the proposed method with advanced deep learning techniques, such as attention mechanisms or hybrid models combined

with convolutional neural networks, to further improve the segmentation accuracy and classification results [41–45].

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Declarations

Conflict of interest Not Applicable

Research Involving Human and/or Animals This study involves research using human data (mammography images).

Informed Consent Not Applicable

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